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School Satisfaction Predicts Quality of Life for Children With Severe Developmental Disabilities and Their Families

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ABSTRACT

Background: Children with severe developmental disabilities are frequently excluded from research, and little is known about their quality of life (QoL). Using a mixed-methods approach, this study examined relationships between school factors and QoL for these children and their families.

Method: 171 parents of children with severe developmental disabilities completed questionnaires. Hierarchical regression analyses were performed examining predictors of child and family QoL. Of the 171 parents, 123 responded to an open-ended question about their children's school experiences, and responses were analysed qualitatively.

Results: Significant predictors of QoL included challenging behaviours, diagnoses, parent self-efficacy, social support and (importantly) school satisfaction. Seven themes related to school experiences were identified qualitatively.

Conclusion: Many factors contribute to QoL. School has a significant influence on children and their parents and families. Different children have different strengths and difficulties, and school systems need to work with parents to optimise outcomes.

1 | Background

This study sought to examine the relationships between school-related factors and quality of life (QoL) for children with severe developmental disabilities and their families, taking an Ecological Systems Theory approach (Bronfenbrenner 1979). Previous research on QoL for these children has been minimal, and the relationship between school factors and QoL remains unclear.

Developmental disability encompasses lifelong cognitive and adaptive functioning difficulties often linked to neurological or physical impairments (Sigafoos et al. 2020). More severe developmental disabilities are associated with more extensive support needs, challenging behaviours (e.g., aggression, self-injurious behaviours), limited communication, and co-occurring conditions (e.g., autism, Down syndrome, cerebral palsy, mental health problems) complicating access to education, living accommodations and socialisation (Matson and Rivet 2008; Simões et al. 2016). These challenges may also contribute to caregiver and familial distress (Berg et al. 2019).

Nonetheless, children with severe developmental disabilities and their families can also experience positive outcomes. Parents report personal growth, greater understanding and acceptance of differences, enriched family relationships and greater appreciation of life (Beighton and Wills 2017; King et al. 2012). These benefits are often seen as direct results of challenges related to the child's disability. Evidently, severe developmental disabilities are incredibly complex, and further research could enhance understanding and support for those impacted.

School is often a focal point in the lives of children with severe developmental disabilities and is where they learn skills, develop relationships and build independence. Inclusive education policies aim to integrate students requiring special education programming into classrooms alongside their typically developing peers (referred to as 'regular classrooms' in Ontario; Ontario Ministry of Education 2014). While well-intentioned, these policies are not necessarily effective (Bhopti, Brown, and Lentin 2020; Kurth, Love, and Pirtle 2020; Perry et al. 2020; Webster and Carter 2013). Effective inclusion has many benefits (Carrington et al. 2016; Ruijs and Peetsma 2009; Simón et al. 2022; Webster and Carter 2013), but it should not be limited to a child's physical presence in a regular classroom. Individual education plans (IEPs) and collaboration between educational staff and other professionals (e.g., psychologists, speech and language pathologists, occupational therapists) may be required (Sigafoos et al. 2020). The success of inclusion relies on the knowledge, expertise and resources of the educational staff (Dillenburger 2012). Unfortunately, many teachers feel ill-equipped to implement inclusion effectively for children with severe developmental disabilities (Carrington et al. 2016). While some children's needs can be met by additional support from general education teachers within regular classrooms, others may require educational assistants, specialised supports and alternative school placements (Dillenburger 2012).

Parents of children with severe developmental disabilities report greater support from specialised school programmes (Bhopti, Brown, and Lentin 2020) and lower satisfaction with regular classrooms (Perry et al. 2020). Many parents report that engaging with schools and support services is their greatest challenge (Kerr, Sharry, and Wilson 2023), citing a lack of agency in educational decisions and difficulties accessing assistive services within schools (Kurth, Love, and Pirtle 2020). Satisfaction with schools tends to be low, but experiences are highly variable, and satisfaction can fluctuate across time and circumstance (Kurth, Love, and Pirtle 2020).

Many factors influence parental school satisfaction. More severe disability symptoms (e.g., social difficulties, negative emotionality, challenging behaviours) are associated with lower satisfaction. Parents of children with more severe symptoms tend to have higher expectations for specialised assistance and staff qualification, training and skill (Vohra et al. 2014). Co-occurring conditions also contribute to parental dissatisfaction (Zablotsky, Boswell, and Smith 2012) making the high prevalence of co-occurring conditions among individuals with severe developmental disabilities a concern (Einfeld, Ellis, and Emerson 2011; El Mrayyan, Eberhard, and Ahlström 2019; Sigafoos et al. 2020). Children with higher adaptive behaviour levels show faster academic improvement and adjustment to school settings (Schneider et al. 2022). Parental positivity, child adaptive skill level and classroom placement type also significantly predict school satisfaction (Perry et al. 2020). The challenges that children experience in adjusting, the challenges that parents face in supporting children and the capacity of the school systems to provide necessary intensive supports all impact parental satisfaction with school experiences. It stands to reason that this satisfaction (or lack thereof) may impact general wellbeing for these families.

QoL assesses individuals' overall wellbeing and includes factors such as material, physical and emotional wellbeing (e.g., personal belonging, healthcare access, self-concept), personal development (e.g., learning opportunities), self-determination (e.g., personal goals), interpersonal relationships (e.g., friendships), social inclusion (e.g., activity participation) and personal rights (e.g., equality; Heras et al. 2021). As such, higher QoL embodies more positive circumstances across essential aspects of life. Predictors of QoL include age (Ncube, Perry, and Weiss 2018; Vos et al. 2010), adaptive skills (Balboni, Mumbardó-Adam, and Coscarelli 2020; Brown et al. 2006; Simões et al. 2016), co-occurring conditions (Leader et al. 2021; Ncube, Perry, and Weiss 2018), caregiver wellbeing (Feng et al. 2022; Lei and Kantor 2022; Ncube, Perry, and Weiss 2018; Wang, Liu, and Zhang 2022), social support (Cai et al. 2023; Hassanein, Adawi, and Johnson 2021; Lei and Kantor 2022; Wang, Liu, and Zhang 2022), socioeconomic status (Knorst et al. 2021), community size (Helliwell, Shiplett, and Barrington-Leigh 2019; Maier and Kim 2008) and school satisfaction (Ncube, Perry, and Weiss 2018). These factors are also relevant to individuals with severe developmental disabilities (Petry, Maes, and Vlaskamp 2005).

The family unit is particularly important for families supporting member(s) with severe developmental disabilities. Family QoL (FQoL) encompasses the wellbeing of these family units. Many factors influence FQoL. Support from others, particularly specialised services and relationships with public service providers, significantly predicts FQoL (Bhopti, Brown, and Lentin 2020; Cai et al. 2023; Hassanein, Adawi, and Johnson 2021; Lei and Kantor 2022; Steel et al. 2011; Wang, Liu, and Zhang 2022). However, school settings and educators are rarely considered among these professional support services, a critical oversight given the significant role of schools in children's lives.

The accessibility of support services varies, with rural communities often having fewer resources (Sibley and Weiner 2011). Yet, adults in urban areas tend to report lower life satisfaction than those in smaller communities (Helliwell, Shiplett, and Barrington-Leigh 2019; Maier and Kim 2008). The impact of increased access to services in urban centres (such as those available within larger schools) on QoL for families of children with severe developmental disabilities needs further investigation.

QoL for individual members of a family should be addressed in addition to FQoL. Research on child QoL (CQoL), especially for children with severe developmental disabilities, is limited. Although these children may have limited capacity to complete self-report measures, their QoL should not be disregarded. Ncube, Perry, and Weiss (2018) conducted a study on CQoL utilising a sample overlapping with the present study. Using parental perceptions, they found that younger age, greater adaptive skills, lower severity of challenging behaviours, greater parent mental health problems and higher school satisfaction predicted higher CQoL. School satisfaction was the strongest individual predictor of CQoL, but their measure was a single questionnaire item, highlighting a need for further investigation. Although their finding that overall school satisfaction is predictive of CQoL is important, it would be beneficial to examine a more thorough measure of school satisfaction.

Direct focus on QoL within the education system could improve wellbeing for children with severe developmental disabilities and their families, but the relationships between CQoL, FQoL and school factors require further exploration (Faragher and Van Ommen 2017; Heras et al. 2021). Children with severe developmental disabilities require unique educational planning, and the ways in which this is executed may impact their lives and their families' lives. The purpose of the present study was to explore parents' perceptions of the QoL of their children with severe developmental disabilities and to examine predictors of QoL for these children and their families. Specifically, we sought to examine school-related experiences and school satisfaction in relation to QoL, addressing the gaps in existing research and providing insights for better support and understanding of families impacted by severe developmental disabilities.

2 | Method

2.1 | Design

Data for this study were collected from 2010 to 2012 as part of the team Great Outcomes for Kids Impacted by Severe Developmental Disabilities (GO4KIDDS) project which included several survey and in-person studies. Ethics approval was obtained from the host university. The present study implemented a secondary analysis of the data from the survey project, taking a mixed-methods approach.

2.2 | Participants

Participants included a convenience sample of parents/caregivers of school-aged children with severe developmental disabilities. They were recruited through invitations distributed to 512 health-based and social service providers across Canada and through online advertisements to Canadian parent organisations. In recruitment materials, "severe developmental disability" was described as an intellectual disability in the moderate-to-profound range, with or without additional diagnoses including autism, physical disabilities, genetic conditions and behavioural or mental health problems. The present study included 171 caregivers (93.0% female) ranging in age from 24 to 58 years ($M = 42.38$; Table 1). Children with severe developmental disabilities were 68.8% male ($n = 117$) and ranged in age from 4 to 19 years ($M = 11.03$). All participants resided in Canada. Most children attended public schools with 11.8% in regular classrooms with minimal assistive services, 35.5% in regular classrooms with one-to-one assistance and 52.7% in special education classrooms.

TABLE 1 | Sample characteristics.

	N	%
Child		
Sex		
Male	117	68.8
Female	53	31.2
Diagnosis		
Severe developmental disability only	74	43.5
Severe developmental disability and autism	96	56.5
School		
Public	152	88.9
Private	14	8.2
Other	6	3.0
Classroom type		
Regular class, no full-time support	20	11.8
Regular class, full-time 1:1 support	60	35.5
Special education class	89	52.7
Community size		
Urban	74	43.5
Suburban	54	31.8
Rural or remote	42	24.7
Parent/family		
Sex		
Male	12	7.0
Female	159	93.0
Country of birth		
Canada	130	76.0
Other	41	24.0
Education		
Less than high school	6	3.6
High school	27	15.9
Some college	25	14.7
College or university graduate	86	50.6
Graduate degree	26	15.3
First language		
English	143	83.6
Other	28	16.4

2.3 | Measures

Data were derived from parent surveys conducted through the GO4KIDDS project (Perry and Weiss 2008).

2.3.1 | CQoL

Parents' perception of CQoL was measured using a 3-item scale from the GO4KIDDS survey (Perry and Weiss 2008). The three items ask parents to rate the quality of their child's friendships, how well they believe their child is achieving their potential and how happy their child seems compared to other children the same age (Ncube, Perry, and Weiss 2018). Each item is rated on a 5-point Likert scale. These ratings are summed to compute a continuous score with potential values ranging from 3 to 15. The established scale reliability reported by Ncube, Perry, and Weiss (2018) for the composite CQoL score is acceptable for a scale of this length ($\alpha = 0.64$). Reliability in our sample was weaker ($\alpha = 0.42$). Notably, Cronbach alpha values are sensitive to the number of items in a scale with low values common among scales with fewer than ten items. Interitem correlation between friendship and achievement was moderate ($r = 0.32$), with weaker correlations observed between friendship and happiness ($r = 0.16$), and achievement and happiness ($r = 0.10$). The total score had strong correlations with several of our other measures.

2.3.2 | FQoL

FQoL was measured using the Beach Center Family Quality of Life Scale (Hoffman et al. 2006). This measure is made up of 25 total items divided across five subscales: family interaction, parenting, emotional wellbeing, physical and material wellbeing, and disability-related support. Each item is rated on a Likert scale from 1 (*very dissatisfied*) to 5 (*very satisfied*). Scores can be computed for individual subscales and a total FQoL score. Internal consistency in our sample was excellent ($\alpha = 0.94$).

2.3.3 | School Services

Parents were asked to identify whether their children received any of five specialised services through their schools: speech and language therapy, occupational therapy, physiotherapy, behaviour therapy and problem behaviour support. Parents also indicated whether each service was a direct intervention or consultation only. Two points were allocated for each direct intervention service and one point was allocated for each consultation service. Points were summed to compute total scores ranging from 0 (no services received from schools) to 10 (direct interventions received across all five school services listed).

2.3.4 | School Satisfaction

Parent satisfaction with schools was measured with the GO4KIDDS School Satisfaction Scale (Perry et al. 2019). The 9-item scale measures parent satisfaction with their child's type of classroom and education programme, classroom staff, communication with school staff, specialised treatments received in school, process of setting goals by the school, academic progress, behavioural progress, social skills progress, inclusion in class activities and social inclusion by typically developing peers. Each item is rated on a Likert scale ranging from 1 (*very dissatisfied*) to 5 (*very satisfied*). The scale demonstrated high internal consistency in our sample ($\alpha = 0.93$).

2.3.5 | Additional Measures

2.3.5.1 | Child Behaviour. Adaptive and challenging behaviours were measured using the Scales of Independent Behaviour-Revised (Bruininks et al. 1996). Ratings for the 35 adaptive items (e.g., motor skills, communication, self-care) ranging from 0 (*never/rarely does*) to 3 (*does very well*). In our sample, the adaptive scale had excellent internal consistency ($\alpha = 0.96$). The challenging behaviour scale consists of eight behaviours (e.g., aggression, inattention, noncooperation) which are rated based on frequency and severity. Scores can range from the most severe challenging behaviours (−72) to the average range (−10 to +10). This scale also had strong internal consistency in our sample ($\alpha = 0.92$).

2.3.5.2 | Parent Mental Health. Parental mental health difficulties (e.g., nervousness, hopelessness, sadness) were measured with the Kessler Psychological Distress Scale (K6; Kessler et al. 2003). The six items are rated on 5-point Likert scale. Internal consistency in our sample was strong ($\alpha = 0.90$).

2.3.5.3 | Parent Self-Efficacy. Parenting self-efficacy was measured using the Family subscale of the Family Empowerment Scale (Koren, DeChillo, and Friesen 1992). The 12 items (related to confidence in parenting and feelings about the family unit) are rated on a 5-point Likert scale. Internal consistency in our sample was strong ($\alpha = 0.90$).

2.3.5.4 | Socioeconomic Status. Socioeconomic status was measured using a modified version of the Barratt Simplified Measure of Social Status (Barratt 2005). Two raters assigned scores to the respondent's (i.e., caregiver) and respondent's partner's (if applicable) highest level of schooling completed (ranging from 3 to 21) and occupation (ranging from 5 to 45). Ratings were combined to compute a continuous total score. Interrater reliability was strong with a correlation of $r = 0.93$.

2.3.5.5 | Social Support. Social support was measured using a modification of a brief scale from the GO4KIDDS Basic Survey (Perry and Weiss 2008). Respondents are asked to rate the perceived helpfulness of three sources of family support (their own parents, in-laws and extended family) and four sources of community support (friends, neighbours, coworkers and religious or cultural groups). Items are rated on 5-point Likert scales ranging from −1 (*make it more difficult*) to 3 (*extremely helpful, I depend on them*). Ratings are summed to compute two subscale total scores (i.e., for family and community support).

2.3.5.6 | Other Variables. Additional demographic information that was collected included child age, sex and diagnosis (i.e., severe developmental disability alone or severe developmental disability and autism), parent age and sex, type of classroom placement (i.e., regular classroom without one-to-one support, regular classroom with one-to-one support or specialised classroom) and community size (i.e., rural/remote, suburban or urban).

Parents were also invited to provide written responses to an open-ended question ("please add any comments about your child's school situation”) with the goal of better understanding school experiences and aspects of school satisfaction not addressed sufficiently by survey measures. Of the total sample of 171 parents, 123 provided written responses.

2.4 | Analyses

2.4.1 | Quantitative Analyses

Data were analysed using IBM SPSS Statistics V28 software (IBM Corp 2021). Preliminary analyses were completed to examine the ranges and distributions of data to ensure no assumptions were violated. Internal consistencies were examined for outcome variables and predictors. Correlational analyses were computed to understand relationships between variables. Two four-step hierarchical regressions were conducted to examine predictors of CQoL and FQoL (Flora 2018).

2.4.2 | Qualitative Analyses

There are limitations to using scales from questionnaires to predict QoL. This study utilised a convergent design for mixed-methods research whereby qualitative and quantitative analyses were combined to compare and expand upon results (Leko et al. 2023). Specifically, we qualitatively analysed written comments about school experiences to broaden our understanding of the relationships between school factors and QoL for our participants. Themes related to school experiences were established utilising a latent, theoretical thematic analysis method (Braun and Clarke 2006). To understand which aspects of school experiences attracted certain emotions, a human-coded sentiment analysis method was used to identify responses as positive, negative or neutral (Thelwall 2022). A primary coder (the first author) compiled preliminary impressions of the data, searched for themes, developed operational definitions for them and named the themes according to Braun and Clarke (2006)'s recommendation. Two coders then assessed the responses separately and assigned them to thematic categories and identified emotional sentiments in responses. Intercoder agreement was strong at 86.0%.

3 | Results

Descriptive statistics are presented in Table 2. Correlations were calculated to examine the relationships of CQoL and FQoL with all other variables (Table 3). Correlations of $|r| \geq 0.50$ were considered strong and correlations of $|r| \geq 0.30$ were considered moderate (Cohen 1988). Both CQoL and FQoL had moderate correlations with child-challenging behaviours, parent mental health, family social support and school satisfaction. CQoL was also moderately correlated with parent self-efficacy while FQoL was strongly correlated with parent self-efficacy. Parent FQoL and parent perception of CQoL were correlated moderately ($r = 0.46$).

3.1 | CQoL

The main analyses included measures of variables related to children, parents and families, communities, and schools, based on Ecological Systems Theory. The first hierarchical regression examined predictors of CQoL (Table 4). In the final model, the significant predictors were more severe challenging behaviours ($\beta = 0.240$, $p < 0.01$), diagnosis of autism ($\beta = -0.176$, $p < 0.05$), family social support ($\beta = 0.183$, $p < 0.05$), community social support ($\beta = 0.159$, $p < 0.05$) and school satisfaction ($\beta = 0.225$, $p < 0.01$). The overall model accounted for 37.5% of the variance in CQoL.

3.2 | FQoL

The second hierarchical regression regarded predictors of FQoL (Table 5). In the final model, the significant predictors were more severe challenging behaviours ($\beta = 0.177$, $p < 0.05$), parent self-efficacy ($\beta = 0.442$, $p < 0.01$), family social support

TABLE 2 | Descriptive statistics for continuous variables.

Measure	*M*	SD	Minimum	Maximum	Possible range
FQoL	3.67	0.67	1.48	5.00	1 to 5
CQoL	8.05	2.21	3.00	14.00	3 to 15
Adaptive behaviour	55.00	20.67	3.00	99.00	0 to 105
Challenging behaviour	−15.57	12.58	−56.00	4.00	−72 to 10
Parent mental health	6.47	5.26	0.00	24.00	0 to 24
Parent self-efficacy	3.87	0.59	2.33	5.00	1 to 5
Socioeconomic status	38.63	14.58	8.00	66.00	8 to 66
Family support	1.99	2.09	−3.00	7.00	−3 to 7
Community support	1.94	1.90	−4.00	10.00	−4 to 10
School services	2.77	2.06	0.00	10.00	0 to 10
School satisfaction	3.53	1.05	1.00	5.00	1 to 5

TABLE 3 | Correlation matrix.

Variable	FQoL	CQoL	Sex	Age	Dx	CHL	ADP	PMH	PSE	FSS	SES	COM	CSS	CRM	SRV
CQoL	0.46*	—													
Sex	0.07	0.13	—												
Age	−0.09	−0.11	0.06	—											
Dx	−0.16	−0.28	−0.45*	−0.18	—										
CHL	0.41*	0.41*	0.11	−0.09	−0.23	—									
ADP	0.16	0.05	−0.06	0.19	0.06	−0.09	—								
PMH	−0.47*	−0.30*	−0.11	−0.04	0.28	−0.44*	−0.10	—							
PSE	0.61**	0.30*	0.09	−0.11	−0.11	0.26	0.05	−0.40*	—						
FSS	0.33*	0.31*	0.02	−0.16	−0.06	0.12	0.09	−0.18	0.24	—					
SES	−0.13	−0.01	0.08	0.07	0.03	−0.09	0.15	0.07	−0.14	−0.09	—				
COM	−0.12	−0.02	0.01	0.04	0.07	0.01	0.10	0.07	−0.01	−0.01	0.12	—			
CSS	0.23	0.27	−0.11	−0.18	−0.05	0.16	0.11	−0.31*	0.13	0.10	0.06	−0.06	—		
CRM	−0.15	−0.10	−0.00	0.25	0.05	−0.05	−0.17	0.00	−0.12	−0.07	−0.01	0.15	−0.22	—	
SRV	0.22	0.21	−0.04	−0.16	−0.13	0.19	−0.22	−0.19	0.13	0.17	−0.12	−0.21	0.06	0.04	—
SAT	0.36*	0.37*	0.13	−0.08	−0.12	0.26	0.10	−0.28	0.14	0.17	−0.08	0.03	0.11	0.11	0.19

Abbreviations: ADP = adaptive behaviour, CHL = challenging behaviour, COM = community size, CRM = classroom type, CSS = community social support, Dx = diagnosis, FSS = family social support, PMH = parent mental health, PSE = parent self-efficacy, SAT = school satisfaction, SES = socioeconomic status, SRV = school services.
*Moderate correlation ($|r| \geq 0.30$).
**Strong correlation ($|r| \geq 0.50$).

TABLE 4 | Regression predicting child quality of life.

Model		*B*	SE *B*	*β*	*t*	R^2	ΔR^2
1	(Constant)	11.014	0.930	—	11.846**	0.228	—
	Challenging behaviours	0.061	0.014	0.348	4.354**		
	Adaptive behaviours	0.012	0.008	0.115	1.460		
	Age	−0.097	0.054	−0.144	−1.783		
	Diagnosis	−1.038	0.361	−0.234	−2.878**		
2	(Constant)	8.224	1.623	—	5.067**	0.298	0.070
	Challenging behaviours	0.051	0.015	0.290	3.408**		
	Adaptive behaviours	0.008	0.008	0.071	0.909		
	Age	−0.060	0.054	−0.089	−1.114		
	Diagnosis	−0.897	0.356	−0.202	−2.519*		
	Parent mental health	−0.009	0.038	−0.022	−0.240		
	Parent self-efficacy	0.485	0.310	0.130	1.566		
	Family social support	0.222	0.083	0.210	2.687*		
3	(Constant)	7.519	1.679	—	4.478**	0.323	0.025
	Challenging behaviours	0.050	0.015	0.285	3.381**		
	Adaptive behaviours	0.005	0.008	0.050	0.641		
	Age	−0.035	0.054	−0.052	−0.649		
	Diagnosis	−0.890	0.353	−0.200	−2.522*		
	Parent mental health	0.012	0.039	0.029	0.312		
	Parent self-efficacy	0.499	0.307	0.134	1.626		
	Family social support	0.222	0.082	0.210	2.709**		
	Community size	−0.018	0.204	−0.007	−0.088		
	Community social support	0.197	0.092	0.170	2.133*		
4	(Constant)	4.976	1.836	—	2.711**	0.375	0.052
	Challenging behaviours	0.042	0.015	0.240	2.879**		
	Adaptive behaviours	0.003	0.008	0.026	0.333		
	Age	−0.004	0.056	−0.006	−0.070		
	Diagnosis	−0.783	0.348	−0.176	−2.256*		
	Parent mental health	0.027	0.039	0.065	0.701		
	Parent self-efficacy	0.475	0.300	0.128	1.581		
	Family social support	0.193	0.081	0.183	2.393*		
	Community size	0.034	0.208	0.013	0.165		
	Community social support	0.184	0.092	0.159	2.007*		
	Classroom type	0.315	0.359	0.071	0.879		
	School services	0.069	0.085	0.065	0.814		
	School satisfaction	0.472	0.163	0.225	2.895**		

*$p < 0.05$.
**$p < 0.01$.

TABLE 5 | Regression predicting family quality of life.

Model		*B*	SE *B*	*β*	*t*	R^2	ΔR^2
1	(Constant)	4.078	0.283	—	14.408**	0.221	—
	Challenging behaviours	0.021	0.004	0.391	4.864**		
	Adaptive behaviours	0.007	0.003	0.223	2.811**		
	Age	−0.024	0.017	−0.117	−1.442		
	Diagnosis	−0.137	0.110	−0.101	−1.239		
2	(Constant)	1.825	0.410	—	4.445**	0.511	0.290
	Challenging behaviours	0.011	0.004	0.207	2.913**		
	Adaptive behaviours	0.004	0.002	0.136	2.093*		
	Age	−0.007	0.014	−0.036	−0.543		
	Diagnosis	0.029	0.090	−0.022	−0.322		
	Parent mental health	−0.021	0.010	−0.162	−2.146*		
	Parent self-efficacy	0.502	0.078	0.445	6.405**		
	Family social support	0.049	0.021	0.152	2.326*		
3	(Constant)	1.915	0.425	—	4.503**	0.527	0.016
	Challenging behaviours	0.011	0.004	0.215	3.044**		
	Adaptive behaviours	0.005	0.002	0.142	2.178*		
	Age	−0.004	0.014	−0.021	−0.310		
	Diagnosis	−0.020	0.089	−0.015	−0.220		
	Parent mental health	−0.017	0.010	−0.135	−1.738		
	Parent self-efficacy	0.505	0.078	0.448	6.500**		
	Family social support	0.048	0.021	0.151	2.340*		
	Community size	−0.097	0.052	−0.117	−1.874		
	Community social support	0.018	0.023	0.050	0.755		
4	(Constant)	1.276	0.466	—	2.741**	0.562	0.036
	Challenging behaviours	0.009	0.004	0.177	2.541*		
	Adaptive behaviours	0.004	0.002	0.124	1.864		
	Age	0.004	0.014	0.018	0.254		
	Diagnosis	0.007	0.088	0.005	0.082		
	Parent mental health	−0.013	0.010	−0.105	−1.361		
	Parent self-efficacy	0.499	0.076	0.442	6.551**		
	Family social support	0.041	0.020	0.129	2.015*		
	Community size	−0.083	0.053	−0.101	−1.580		
	Community social support	0.015	0.023	0.041	0.624		
	Classroom type	0.078	0.091	0.058	0.853		
	School services	0.019	0.022	0.057	0.861		
	School satisfaction	0.117	0.041	0.184	2.831**		

*$p < 0.05$.
**$p < 0.01$.

($\beta=0.129$, $p<0.05$) and school satisfaction ($\beta=0.184$, $p<0.01$). The overall model accounted for 56.2% of the variance in FQoL.

3.3 | Qualitative Results

Seven themes related to school experiences were identified. Overall sentiments were 54.1% negative, 27.6% positive and 18.3% neutral (see Table 6 for sentiment breakdown by theme).

3.3.1 | Themes

3.3.1.1 | Salient Personal Experiences. This theme consisted of overall experiences and significant, personal experiences within the school system. Sentiments in this theme were primarily negative and responses tended to be specific, complex and extremely salient. For example, one mother spoke about her son in a special education class saying, "my child has 2:1 assistance at all times and escaped from school. He was found running down a busy street, and not by school staff. It is very stressful leaving him with other caregivers knowing the dangers." Despite of the overwhelming negative sentiments in this theme, there were also positive accounts, with another mother indicating, 'our school has been exceptional, and it is a main reason why we have stayed in this community'. Few responses were identified as neutral, mentioning personal experiences without clear emotional undertones, such as the mother who indicated, 'I do not [know] the other kids in his class or the other families'.

3.3.1.2 | School Staff. This theme was in relation to communication and relationships with educational staff (including teachers, support staff and administrative staff). Sentiments were fairly evenly divided between positive and negative with a slightly higher occurrence of negative responses. In reference to her son's special education programme, one mother described a 'totally inexperienced teacher, lazy, irresponsible. No passion in kids, not to say kids with special needs. Can't even do an IEP right. Refused to implement recommendations by [speech and language pathologist] saying it's too much'. However, many caregivers identified positive experiences with members of their children's school staff. A mother of a young boy in a regular kindergarten classroom suggested that 'classroom teacher and [educational assistant] are wonderful to work with and have built an amazing relationship with student and home'. Neutral responses tended to include more generalised statements about educational personnel such as the mother who explained that her 'child's academic level directly depends on the teacher's ability to tolerate his disability and how well the teacher and school staff are educated about disability and different strategies to deal with him'.

3.3.1.3 | The Education System. This theme related to the performance of the education system, in general. The majority of sentiments in this theme were negative, indicating many frustrations with the school system. A mother of a boy in a regular classroom with one-to-one assistance explained that 'school has been even more challenging than the disability itself. The IEP process is a joke and the lack of knowledge on how to meaningfully educate a child with profound challenges is incredibly frustrating'. Responses that were positive often included elements of frustration with systems beyond the school. For example, one father indicated 'the school division has done very well for the limited resources they are given'. Neutral responses tended to describe how parents were interacting with the system. Describing her 13-year-old son, one mother indicated, 'at this point in time we have taken the reins where our child's IEP is concerned, we have now re-directed the team towards life skills and preparation for impending adulthood'.

3.3.1.4 | Educational Placements. This theme encompassed the different types of educational placements (classroom, school, educational support, etc.) that children experienced. Over half of the responses to this theme were neutral, stating what sorts of placements children were in. Regarding her 12-year-old son, one mother simply stated, 'he is in a regular classroom with one-on-one support'. There were equal numbers of positive and negative responses. Where one mother indicated, 'my son is not ready for a regular classroom, so I am very, very grateful to have a specialized class to meet his needs', another specified that 'although he is supposed to be integrated in a regular classroom, I have recently learned that he is spending most of his time in the learning centre, thus missing out on social connections with his own peer group'.

3.3.1.5 | Temporal Changes. This theme was in reference to changes that occurred over time as well as the unpredictability and instability that occurred as a result. The most common sentiment in this theme was neutral, with one mother explaining that 'there is huge variation between years, teachers, and schools'. Negative responses tended to allude to fear or apprehension about the future as was the case with a mother who described, 'my daughter will be moving on to high school next year where I feel she will not get the same education and will just be "babysat"'. Positive responses often mentioned previous dissatisfaction but success in the present. Regarding her 16-year-old son's special education class, one mother stated, 'the school situation has changed radically this fall—he has a teacher who has expectations and is willing to push him. The previous 2 years of schooling were entirely inadequate, but for the moment we are satisfied'.

3.3.1.6 | Inclusion. This theme regarded inclusion and peer relationships. Over half of the responses to this theme had negative sentiments such as with a mother who explained that 'in the public they will promote inclusion, but that is not what happens out of the public's eye'. Positive responses generally included praise for how children were included within schools. The mother of a 14-year-old student in a regular classroom with one-to-one assistance stated, 'students are supported to include her and treat her kindly, and many people are able to look beyond the challenging behaviours and appreciate my daughter's gifts'. Neutral responses simply stated the presence or absence of inclusion, including a mother who indicated, '[my child is] getting plugged in with inclusion'.

3.3.1.7 | Learning, Achievement and Growth. This theme referred to what a child learned in school and the extent of their success in achieving goals set by their family and the school. The majority of responses of this theme were negative. One father expressed frustration with his son's special education programme, stating, 'they are not educating our son.

TABLE 6 | Frequency of sentiments by theme.

Theme	% positive	% negative	% neutral
Salient personal experiences	28.6	70.5	1.0
School staff	41.6	51.5	6.9
The education system	12.2	80.5	7.3
Educational placements	23.5	23.5	53.1
Temporal changes	25.3	36.0	38.7
Inclusion	32.7	51.7	15.5
Learning, achievement, and growth	26.5	61.2	12.2

He is being babysat for his most vital learning hours'. Positive responses typically referred to personal growth and achievement, with one mother describing that her daughter 'has already grown so much and tells us what she is learning which she never did before'. Neutral responses tended to describe what goals children were working towards. For example, one mother described that 'the students work on both academic skills and social skills' in her daughter's special education class.

4 | Discussion

The purpose of this mixed-methods study was to better understand the relationships between school factors and QoL for children with severe developmental disabilities and their families. This was investigated through the analysis of several questionnaire measures and written accounts of school experiences provided by caregivers. Variables were considered at the levels of the child, the parents and family, the community and the school system. Significant predictors of CQoL included challenging behaviours, diagnoses, family social support, community social support and school satisfaction. Significant predictors of FQoL included challenging behaviours, parent self-efficacy, family social support and school satisfaction. Complementing the findings on school satisfaction, seven themes were identified based on parents' written comments related to school experiences.

4.1 | Child Factors

Aligned with an Ecological Systems Theory perspective, CQoL and FQoL were associated with several concepts throughout all levels of the environment. The child-specific variables that we considered were adaptive and challenging behaviours, sex, age and diagnosis. As has been found in previous research, increased challenging behaviours in our sample predicted lower QoL for both children and their families (Brown et al. 2006; Ncube, Perry, and Weiss 2018). Challenging behaviours such as aggression, self-injury and noncompliance can result in fear and frustration from others which may limit opportunities for both children and their parents to have positive social interactions and relationships. Additionally, the motivation behind these behaviours may be related to different aspects of (un)happiness and wellbeing (e.g., the child may be in pain or frustrated that they cannot effectively communicate).

Interestingly, we did not find a similar relationship between adaptive behaviours and QoL. This is contrary to previous research on adults with developmental disabilities, which has found that higher levels of adaptive behaviours do predict increased individual QoL (Balboni, Mumbardó-Adam, and Coscarelli 2020; Simões et al. 2016), but consistent with findings from a study on autistic children in which adaptive behaviour was not associated with QoL (Schneider et al. 2022). This could imply that the negative impact of challenging behaviours on both individual children and their families is amplified when considering wellbeing, but age-expected behaviours do not necessarily have a conversely positive impact.

CQoL was significantly predicted by diagnosis meaning that children with severe developmental disabilities alone had higher CQoL than children with severe developmental disabilities and autism. A possible explanation for this finding could be the inclusion of friendship quality in the CQoL measure because this would naturally be expected to be lower in autistic children since a defining characteristic of autism is difficulty with social communication and relationships. Furthermore, autism was mildly associated with challenging behaviour. That is, children with both severe developmental disabilities and autism had slightly increased occurrences of challenging behaviours which may be linked to lower CQoL. This is aligned with previous research (Leader et al. 2021). However, diagnosis was not associated with FQoL, suggesting that there was no difference in QoL for parents of children with only severe developmental disabilities and children with both severe developmental disabilities and autism. It is likely that it is not the diagnosis but the associated symptoms that are particularly important for parents' FQoL. This is reflected by the influence of challenging behaviour.

In addition, we did not find any associations between QoL and child sex or age. While there is little precedence for an association between sex and QoL, some research has found that increased age may be associated with lower QoL (Ncube, Perry, and Weiss 2018; Vos et al. 2010). Subjectively, we observed several written comments that indicated a drop-off of support services as children aged which could explain why other studies have found this association. Though our correlational analysis indicated that the relationship between age and support services was not significant, we only accounted for support services available exclusively through schools so this drop-off may not be reflected in our data.

4.2 | Parent and Family Factors

All family variables included in our regression analyses were moderately to strongly correlated with CQoL and FQoL, but not all of them were significant predictors when considered simultaneously. However, socioeconomic status was not significantly associated with either of our QoL measures and was omitted from our regression analyses. A 2021 meta-analysis found several studies indicating that lower socioeconomic status is associated with lower health-related aspects of QoL (Knorst et al. 2021). Notably, Canada has universal healthcare which could relieve a significant amount of burden associated with accessing services. This could make our sample fundamentally different from those studied in previous research.

Social support from extended family significantly predicted both CQoL and FQoL. While there is very little other research specifically on CQoL, this finding is aligned with several studies showing that social support is an impactful aspect of FQoL and QoL in adults (Cai et al. 2023; Hassanein, Adawi, and Johnson 2021; Lei and Kantor 2022; Wang, Liu, and Zhang 2022). Interestingly, even though parents provided responses on behalf of their children, parent mental health and parent self-efficacy did not predict CQoL. Although these parent variables presumably have strong influence on parents themselves, parents do not perceive the variables as related to their children's wellbeing as captured by our measure of CQoL (i.e., happiness, quality of friendships and achievement of potential).

Parent self-efficacy did significantly predict FQoL. The contribution of both parent self-efficacy and social support to FQoL has been mirrored in previous research (Feng et al. 2022). Our measure of parent self-efficacy overlaps with the parenting subscale of our FQoL measure which may partially explain its significance. Parental coping skills, adaptability and family cohesion (concepts captured by both measures) have previously been found to contribute to FQoL (Lei and Kantor 2022; Wang, Liu, and Zhang 2022). Parents who felt more capable of responding to and meeting the needs of their children, able to help their children learn and grow, and confident in their overall parenting skills reported more positive overall FQoL.

Parent mental health was a significant predictor of FQoL in earlier steps of the regression analysis, but with the addition of all other variables, it became insignificant. This indicates that, while parent mental health is important for FQoL, it is not as important as or overlaps with many of the other factors for which we accounted. While other research has found that parent mental health can significantly predict FQoL (Ncube, Perry, and Weiss 2018), it is possible that the variables we considered better captured the concept than those used in other studies.

4.3 | Community Factors

Although previous research has shown that community size can predict QoL (Helliwell, Shiplett, and Barrington-Leigh 2019; Maier and Kim 2008), we did not find this. Notably, previous research has been limited to adults without developmental disabilities, which could explain the difference in our findings. In line with previous findings (Bhopti, Brown, and Lentin 2020; Sibley and Weiner 2011), written comments from parents in our sample indicated that disability support services (which are presumably more accessible in larger communities) were highly desired. However, several comments also indicated that smaller communities felt supportive and like "family" and perhaps this tight-knit bond makes up for a lack of specialised services. Thus, communities of all sizes have benefits and drawbacks, resulting in community size being a poor predictor of QoL in our sample.

Social support from the broader community was a significant predictor of CQoL but not FQoL. This contrasts with a vast majority of research that has found that social support has great influence on FQoL (Cai et al. 2023; Hassanein, Adawi, and Johnson 2021; Lei and Kantor 2022; Wang, Liu, and Zhang 2022). Of note, other research generally combines social support from family members with that of support from friends, neighbours and broader community members, whereas we divided social support into two categories (for family members and nonfamily members). Seeing as family social support was significant in our analyses and community social support was not, we may have captured a more nuanced picture of social support than previous research studies. It is possible that the significance attributed to social support in other studies also stems primarily from family members. Additionally, we considered not only the presence/availability of social support but the perceived helpfulness of these sources. The significance of community social support for CQoL may be a result of the positive benefits a greater support system offers through additional opportunities for recreation participation. However, it is unclear why this additional support is not predictive of FQoL. It may be that because the other variables in our model accounted for more variance in FQoL than community social support, this variable is comparably insignificant.

4.4 | School Factors

In contrast to what we anticipated, classroom placement was unrelated to QoL for both children and their families. Previous research has found that parents of children with severe developmental disabilities tend to be more satisfied with specialised classes and schools (Bhopti, Brown, and Lentin 2020; Perry et al. 2020) so it is somewhat surprising that placement did not impact QoL, especially because our sample partially overlaps with that from Perry and colleagues' study (2020). However, they were specifically interested in factors that predicted school satisfaction, whereas we were concerned with factors that predicted QoL. It may be that classroom placement indeed played a role in determining school satisfaction among our participants, but that the degree of satisfaction with these placements (among other aspects of school) was what predicted QoL. Qualitative analyses may help to shed light on this finding. Parents provided equal instances of positive and negative comments referring to educational placements with the majority of these comments being neutral. Additionally, different parents reported both satisfaction and dissatisfaction with similar placements. For example, where one mother described her son's special education class as being a place where 'all the children have needs which is taxing on staff and other students', another indicated that 'the choice of segregated school with children similar to him has been just wonderful'. Thus, it is likely that it is not the placement itself that influences QoL but how effective the placement is in meeting the needs of the child and their family.

Similarly, school services were unrelated to CQoL and FQoL. Although previous research has shown that positive relationships with public service providers can positively impact QoL (Summers et al. 2007), we did not find this to be the case in the context of the school system. It should be noted that we only accounted for a limited number of service providers and only those received through the school system (no other medical, social, or therapeutic services received elsewhere) so there could be others that are more impactful. On the surface, this finding seems to be in contrast to the several written comments we obtained mentioning the desire for more services within schools. However, like our finding with classroom placement, it may be that it is not simply the presence of these services that is associated with QoL, but their alignment with the needs of particular students. As such, it seems that the broader construct of school satisfaction is the most important aspect of school for children with severe developmental disabilities and their families.

Indeed, we found that school satisfaction significantly predicted both CQoL and FQoL, over and above all other variables. As a concept that has not been directly studied, this finding highlights an immensely influential aspect of life for these children and their families. The concept of satisfaction was seen in every qualitative theme we identified. Both positive and negative experiences within the school system were salient and impactful for these families. Consistent with previous literature, schools seem to have the ability to influence several aspects of life for individuals with severe developmental disabilities (Bhopti, Brown, and Lentin 2020; Carrington et al. 2016; Kurth, Love, and Pirtle 2020; Ncube, Perry, and Weiss 2018; Perry et al. 2020; Ruijs and Peetsma 2009; Simón et al. 2022; Webster and Carter 2013). With the right training, resources and support, educational personnel have the capacity to facilitate wonderful experiences, potentially influencing overall QoL, for these children and their families.

4.5 | Limitations and Future Directions

There are some limitations to the present research study that should be considered. First, we utilised caregivers as respondents for children with severe developmental disabilities since these children were unable to complete self-report measures. However, parents may not have a perfectly accurate picture of how their children feel. Additionally, our measure of CQoL is limited in its scope since it was only made up of three items and it did not have strong reliability ($\alpha = 0.42$ in our sample). Interitem correlations can be a stronger indicator of reliability for scales of this length, where correlations between 0.2 and 0.4 are recommended (Briggs and Cheek 1986). While one of our interitem correlations fell within this range and another was approaching it, one was below. This may indicate multidimensionality among the items that we used to represent the construct of CQoL. It is likely that this could be elucidated by a more comprehensive measure; however, one does not currently exist. Furthermore, some research has suggested that families of children with severe developmental disabilities comprise a functionally heterogeneous group, even when they appear demographically homogeneous (Fitzsimmons 2020), resulting in variable QoL measurements. Nevertheless, we feel it is important to include children with severe developmental disabilities in research since they are often neglected. Future research may seek to establish more comprehensive measures and explore other ways to include this population.

Data for this study were collected in 2012 for a previous project. It is possible that parental perceptions of school and disability have since shifted. Specifically, the COVID-19 pandemic was a major life event that our respondents had not yet experienced. We recognise that the pandemic put significant additional stressors on the education system and those interacting with it and our findings may not apply to this dynamic situation. However, these findings are relevant as education systems have transitioned back to in-person learning formats and educators are implementing specific strategies to close the learning gaps related to the pandemic (Hungerford-Kresser, Cummins, and Amaro-Jiménez 2024). It could be beneficial to replicate our study to account for any temporal changes. Additionally, since our data were collected for a previous project with different objectives, our measures were not selected with the present study in mind. Other measures may have better captured the concepts we were interested in, and future research may consider using other measures to replicate our study.

A final limitation of our study is our sample. While recruitment was broad and included over 500 diverse agencies and parent organisations, it was a convenience sample consisting primarily of parents from urban areas with rather high education levels. We recognise that this may not be representative of all children with severe developmental disabilities. However, it is worth noting that our study consisted of a fully Canadian sample with representation across all provinces. Notably, this may also raise challenges since school systems, government funding allocation and educational philosophies can vary across provinces. Continued research with widely representative Canadian samples may help to provide additional insight into these issues.

4.6 | Conclusion and Implications for Practice

Despite some limitations, this study is one of the first to examine the concept of school factors in relation to QoL for children with severe developmental disabilities and their families. Results indicate that school is incredibly important and can have significant positive or negative impacts. It is not simply the type of school placement nor the provision of specialised school services that predict QoL, but rather whether strategies implemented are deemed by parents as meeting the needs of their children. Our findings may be used to guide educational training, practices and policies to support positive experiences for students with severe developmental disabilities. There is not only a single educational strategy that predicts QoL for children or their families. Educational planning should be individualised, taking into account each child's abilities, needs and goals. This should be done in partnership between educational staff and parents to optimise outcomes for children.

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Ethics Statement

Data collection for this project was approved by the York University Office of Research Ethics (certificate number: 2013-050).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Research data are not shared.

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